

2

* Here I will show that $\lambda^2 = l(l+1)$ for $l=0,1,\dots$.
I'm not too sure if this is required or not so just skip everything in this section if not:

~> Starting from the Legendre equation: $x \in [-1, 1]$
 $(1-x^2)y'' - 2xy' + \lambda^2 y = 0$

also written as

$$\frac{d}{dx} \left[(1-x^2) \frac{dy}{dx} \right] + \lambda^2 y = 0$$

• To find a solution we assume

$$y(x) = \sum_{k=0}^{\infty} c_k x^k$$

plugging this in:

$$(1-x^2) \sum_{k=2}^{\infty} k(k-1) c_k x^{k-2} - 2x \sum_{k=1}^{\infty} k c_k x^{k-1} + \lambda^2 \sum_{k=0}^{\infty} c_k x^k = 0$$

• Putting these together:

$$\sum_{k=0}^{\infty} [\lambda^2 - k(k+1)] c_k x^k + \sum_{k=2}^{\infty} k(k-1) c_k x^{k-2} = 0$$

• In the last sum let $\tilde{k} = k-2$

$$\rightarrow \sum_{k=2}^{\infty} k(k-1) c_k x^{k-2} = \sum_{\tilde{k}=0}^{\infty} (\tilde{k}+2)(\tilde{k}+1) c_{\tilde{k}+2} x^{\tilde{k}}$$

now, relabel $\tilde{k} = k$ and plug back into the above

$$\rightarrow \sum_{k=0}^{\infty} [\{\lambda^2 - k(k+1)\} c_k + \{(k+1)(k+2) c_{k+2}\}] x^k = 0$$

• Thus

$$(k+1)(k+2) c_{k+2} + [\lambda^2 - k(k+1)] c_k = 0$$

$$c_{k+2} = - \frac{(\lambda^2 - k(k+1))}{(k+1)(k+2)} c_k$$

• To get an idea of things note as $k \rightarrow \infty$
 $\lim_{k \rightarrow \infty} (-1 \lambda^2 - k(k+1)) ?$

$$k \rightarrow \infty \left\{ \frac{1}{(k+1)(k+2)} \right\} = 1$$

and so in this limit $c_{k+2} \sim c_k$ and must therefore converge to some c_∞ . In order for the series to converge we require $|x| < 1$

$$c_\infty \sum_{k=1}^{\infty} x^k$$

But we need this to also converge for $x = \pm 1$ since $x \in [-1, 1]$. This means the series must terminate at some $n \geq 0, c_n$. A choice one can make is

$$c_n \neq 0, \quad c_{n+2} = 0$$

This means:

$$c_{n+2} = 0 = - \frac{(\lambda^2 - n(n+1))}{(n+1)(n+2)} c_n$$

and so

$$\lambda^2 = n(n+1) \quad n \in \mathbb{Z}$$

* For physics we use $n = l$:

$$\lambda^2 = l(l+1)$$

Plugging this in we arrive at the form of interest:

$$\frac{d}{dx} \left[(1-x^2) \frac{dy}{dx} \right] + l(l+1)y = 0$$

Of course, we want to get this into spherical coords.

To do this we set $x = \cos(\theta)$

(chain rule): $\frac{d}{dx} = \frac{d\theta}{dx} \frac{d}{d\theta}$

to find $\frac{d\theta}{dx}$: $\frac{d}{dx}(x) = \frac{d}{dx}(\cos(\theta))$

$$1 = -\sin(\theta) \frac{d\theta}{dx}$$

or $\frac{d\theta}{dx} = - \frac{1}{\sin(\theta)}$

• $\frac{d}{dx} = \left[\frac{-1}{\sin(\theta)} \right] \frac{d}{d\theta}$

• Next note that $1-x^2 = 1 - \cos^2(\theta) = \sin^2(\theta)$

$$\frac{d}{dx} \left((1-x^2) \frac{dy}{dx} \right) + l(l+1)y = 0$$

$$\hookrightarrow \left(-\frac{1}{\sin(\theta)} \frac{d}{d\theta} \right) \left[\sin^2(\theta) \left(\frac{-1}{\sin(\theta)} \frac{d}{d\theta} y \right) \right] + l(l+1)y = 0$$

leaving,

$$\frac{1}{\sin(\theta)} \frac{d}{d\theta} \left[\sin(\theta) \frac{dY}{d\theta} \right] + l(l+1)Y = 0$$

for us we have $y = \Theta$:

$$-\frac{1}{\sin(\theta)} \frac{d}{d\theta} \left[\sin(\theta) \frac{d\Theta}{d\theta} \right] = l(l+1)Y$$

$\leadsto$ which is the equation in (1) with $\lambda^2 = l(l+1)$
 $l = 0, 1, \dots$

*What was the point of the above? Well, we can now freely use the equations given in the formula sheet.

Note that Θ satisfies the eigenvalue equation (10)

$$\mathcal{L}_p P_l(x) = l(l+1) P_l(x)$$

with $x = \cos(\theta)$

$$\mathcal{L}_p P_l(\cos(\theta)) = l(l+1) P_l(\cos(\theta))$$

• Note on the operator $\mathcal{L}_p$:

$$\mathcal{L}_p f = -\frac{d}{dx} \left((1-x^2) \frac{df}{dx} \right)$$

which sometimes confuses students

Note that for any given l there is a different solⁿ and hence by superposition the sum of each is as well

$$V(r, \theta) = \sum_l R_l(r) P_l(\cos(\theta))$$

Now we must solve for $R_l(r)$:

$$R'' + \frac{2}{r} R' - \frac{l(l+1)}{r^2} R = 0$$

rewrite as

$$r^2 R'' + 2r R' - l(l+1) R = 0$$

Let $r = e^p$:

$$r^2 = e^{2p}$$

$$\left[\begin{array}{l} \frac{d}{dr} = \frac{dp}{dr} \frac{d}{dp}, \text{ to find } \frac{dp}{dr} : \frac{d}{dr}(r) = \frac{d}{dr}(e^p) \\ 1 = e^p \frac{dp}{dr} \end{array} \right.$$

$$\left[\begin{array}{l} \frac{dp}{dr} = e^{-p} \\ \rightarrow \frac{d}{dr} = e^{-p} \frac{d}{dp} \end{array} \right.$$

• We also need $\frac{d^2}{dr^2}$

$$\frac{d^2 R}{dr^2} = \frac{d}{dr} \frac{dR}{dr} = \frac{d}{dr} \left(e^{-p} \frac{dR}{dp} \right) = e^{-p} \frac{d}{dp} \left(e^{-p} \frac{dR}{dp} \right)$$

$$= e^{-p} \left[-e^{-p} \frac{dR}{dp} + e^{-p} \frac{d^2 R}{dp^2} \right]$$

$$\frac{d^2 R}{dr^2} = e^{-2p} \frac{d^2 R}{dp^2} - e^{-2p} \frac{dR}{dp}$$

• Plugging this all in:

$$r^2 R'' + 2r R' - l(l+1)R = 0$$

$$(e^{2p}) \left[e^{-2p} \frac{d^2 R}{dp^2} - e^{-2p} \frac{dR}{dp} \right] + 2(e^p) \left[e^{-p} \frac{dR}{dp} \right] - l(l+1)R = 0$$

$$\frac{d^2 R}{dp^2} + \frac{dR}{dp} - l(l+1)R = 0$$

• This has solutions of the form $R(p) = e^{\pm ap}$, $a \in \mathbb{R}$
and so $R(p) = A e^{ap} + B e^{-ap}$, A, B, a : constants

Since $r = e^p =$

$$R(r) = A r^a + B r^{-a}$$

In the limit $r \rightarrow 0$ $R(r) \rightarrow \infty$ since $\frac{1}{r^a} \rightarrow \infty$

so for this to be finite this would mean $B=0$.

$$R(r) = A r^a$$

Plugging this into the original ODE

$$r^2 R'' + 2r R' - l(l+1)R$$

$$r^2 a(a-1) A r^{a-2} + 2r A a r^{a-1} - l(l+1) A r^a = 0$$

$$\text{or } a(a-1) + 2a - l(l+1) = 0$$

$$a^2 - a + 2a - l(l+1) = 0$$

$$a^2 + a - l(l+1) = 0$$

$$a(a+1) - l(l+1) = 0$$

$$\text{thus: } \boxed{a(a+1) = l(l+1)}$$

for which $a=l$

Thus

$$R_l(r) = A_l r^l$$

In the problem $A_l = V_l$. Putting this together:

$$V(r, \theta) = \sum_l R_l(r) \Theta_l(\theta) = \sum_l V_l r^l P_l(\cos \theta)$$

$P_l(\cos \theta)$: l^{th} Legendre polynomial (end of ②)

③ Now, we use the Boundary condition at $r=R$

$$V(R, \theta, \phi) = \begin{cases} V_0 & 0 \leq \theta < \frac{\pi}{2} \\ -V_0 & \frac{\pi}{2} < \theta \leq \pi \end{cases}$$

Before we go through the analysis let's swap eq (1) to spherical:

$$\langle P_l, P_{l'} \rangle_{w=1} = \int_{-1}^1 P_l(x) P_{l'}(x) dx$$

$$x = \cos \theta, \text{ thus } \int_{-1}^1 P_l(\cos \theta) P_{l'}(\cos \theta) d(\cos \theta) = \frac{2}{2l+1} \delta_{l,l'}$$

* Now at $r=R$:

$$V(R, \theta, \phi) = \sum_l V_l R^l P_l(\cos \theta)$$

multiply by $P_{l'}(\cos \theta)$

$$P_{l'}(\cos \theta) V(R, \theta, \phi) = \sum_l V_l R^l P_{l'}(\cos \theta) P_l(\cos \theta)$$

integrate

$$\int_{-1}^1 P_{l'}(\cos \theta) V(R, \theta, \phi) d(\cos \theta) = \sum_l V_l R^l \int_{-1}^1 P_{l'}(\cos \theta) P_l(\cos \theta) d(\cos \theta)$$

$$\rightarrow \int_{-1}^1 P_{l'}(\cos \theta) V(R, \theta, \phi) d(\cos \theta) = \sum_l V_l R^l \left(\frac{2}{2l+1} \delta_{l,l'} \right) \frac{2}{2l+1} \delta_{l,l'}$$

the $\delta_{l,l'}$ gets rid of the sum.

$$\int_{-1}^1 P_{l'}(\cos \theta) V(R, \theta, \phi) d(\cos \theta) = V_l R^l \frac{2}{2l+1}$$

LHS: Note $V(R, \theta, \phi) = \begin{cases} V_0, & 0 \leq \theta < \pi/2 \\ -V_0, & \pi/2 < \theta \leq \pi \end{cases}$

Since $x = \cos \theta$

$$\theta = 0: \quad x = 1$$

$$\theta = \pi/2 \quad x = 0$$

$$\theta = \pi \quad x = -1$$

Thus the above can be written as

$$V(R, x, \phi) = \begin{cases} V_0 & 0 \leq x \leq 1 \\ -V_0 & -1 \leq x < 0 \end{cases}$$

and so

$$\int_{-1}^1 P_\ell'(x) V(R, x, \phi) dx = \int_{-1}^0 P_\ell'(x) V(R, x, \phi) dx + \int_0^1 P_\ell'(x) V(R, x, \phi) dx$$

$\uparrow$ $\uparrow$
 $-V_0$ V_0

$$= \int_{-1}^0 P_\ell'(x) (-V_0) dx + \int_0^1 P_\ell'(x) V_0 dx$$

$\underbrace{\hspace{10em}}_{\text{let } x \rightarrow -x}$

$$= - \int_0^1 P_\ell'(-x) V_0 dx + \int_0^1 P_\ell'(x) V_0 dx$$

LHS: $V_0 \int_0^1 (P_\ell'(x) - P_\ell'(-x)) dx$

Plugging this in and setting $\ell' = \ell$ (to match the problem)

$$V_0 \int_0^1 (P_\ell(x) - P_\ell(-x)) dx = V_\ell R^\ell \frac{2}{2\ell+1}$$

Solve for V_ℓ :

$$V_\ell = \frac{V_0 (2\ell+1)}{2 R^\ell} \int_0^1 (P_\ell(x) - P_\ell(-x)) dx$$

define $Y_\ell \equiv \frac{1}{2} \int_0^1 (P_\ell(x) - P_\ell(-x)) dx$

and so

$$V_\ell = V_0 \frac{(2\ell+1)}{R^\ell} Y_\ell$$

* Now using $P_\ell(-x) = (-1)^\ell P_\ell(x)$:

$$\begin{aligned} Y_\ell &= \frac{1}{2} \int_0^1 (P_\ell(x) - (-1)^\ell P_\ell(x)) dx \\ &= \frac{1}{2} [1 - (-1)^\ell] \int_0^1 P_\ell(x) dx \end{aligned}$$

ℓ : even: then $\ell = 2n$, $n \in \mathbb{Z}$

$$[1 - (-1)^{2n}] = [1 - (-1)^{2n}] = [1 - 1] = 0$$

ℓ : odd: $\ell = 2n+1$, $n \in \mathbb{Z}$

$$[1 - (-1)^{2n+1}] = [1 - (-1)^{2n}(-1)] = 2$$

Thus

$$Y_l = \begin{cases} 0, & l \text{ even} \\ \int_0^1 P_l(x) dx, & l \text{ odd} \end{cases}$$

(end of (3))

(4) Plugging this in:

$$x = \cos(\theta)$$

$$V(r, \theta, \phi) = \sum_l V_0 \frac{(2l+1)}{R^l} r^l Y_l P_l(x)$$

from Y_l we know that even l yield $Y_l = 0$ and so the first two terms are $l=1, 3$

$$V(r, \theta, \phi) = V_0 3 \left(\frac{r}{R}\right) Y_1 P_1(x) + V_0 7 \left(\frac{r}{R}\right)^3 Y_3 P_3(x) + \mathcal{O}(r^5)$$

we can read off P_1 & P_3 from (12)

$$P_1 = x, \quad P_3 = \frac{1}{2}(5x^3 - 3x)$$

$$Y_1 = \int_0^1 P_1(x) dx = \int_0^1 x dx = \frac{1}{2}$$

$$Y_3 = \int_0^1 P_3(x) dx = \int_0^1 \frac{1}{2}(5x^3 - 3x) dx = \frac{1}{2} \left[5 \frac{1}{4} - 3 \frac{1}{2} \right]$$

$$Y_3 = -\frac{1}{8}$$

Thus

$$V(r, \theta, \phi) = V_0 3 \left(\frac{r}{R}\right) \left(\frac{1}{2}\right) x + V_0 7 \left(\frac{r}{R}\right)^3 \left(-\frac{1}{8}\right) \left[\frac{1}{2}(5x^3 - 3x)\right]$$

where $x = \cos(\theta)$

(end of (4))

(5) Recall that $\vec{E} = -\nabla V$

with ∇ given by (8)

* Note we want the field at the center meaning at $r=R$. Also there is no ϕ so we can ignore $\frac{\partial}{\partial \phi}$

• Lets do this term by term, (i) $= V_0 \left(\frac{3}{2}\right) \left(\frac{r}{R}\right) \cos(\theta)$

$$(i) -\nabla \left(V_0 \frac{3}{2} \left(\frac{r}{R}\right) \cos(\theta) \right)$$

$$= - \left(\hat{e}_r \frac{\partial}{\partial r} + \hat{e}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} \right) \left(V_0 \frac{3}{2} \left(\frac{r}{R}\right) \cos(\theta) \right)$$

$$= - V_0 \frac{3}{2} \left[\frac{\cos(\theta)}{R} \hat{e}_r - \frac{1}{R} \sin(\theta) \hat{e}_\theta \right]$$

$$= - \frac{V_0}{R} \left(\frac{3}{2}\right) \left[\cos(\theta) \hat{e}_r - \sin(\theta) \hat{e}_\theta \right]$$

This is exactly the basis $\hat{e}_z$

$$\hat{e}_z = \cos(\theta) \hat{e}_r - \sin(\theta) \hat{e}_\theta$$

Thus:

$$t_1 = -\frac{V_0}{R} \left(\frac{3}{2}\right) \hat{e}_z$$

this is the solution as you can see. The second term will vanish after you take derivatives and set $r=0$.

$$\text{Quickly: } -\nabla \left\{ V_0 7 \left(\frac{r}{R}\right)^3 \left(-\frac{1}{8}\right) \left[\frac{1}{2} (5 \cos(\theta))^3 - 3 \cos(\theta) \right] \right\}$$

$$= - \left(\hat{e}_r \frac{\partial}{\partial r} + \hat{e}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} \right) \left[\begin{matrix} \uparrow \\ \end{matrix} \right]$$

$$= - V_0 7 \cdot 3 \frac{r^2}{R^3} \left(-\frac{1}{8}\right) \left[\frac{1}{2} [5 \cos^3(\theta) - 3 \cos(\theta)] \right]$$

$$= - V_0 7 \frac{r^2}{R^3} \left(-\frac{1}{8}\right) \left[\frac{1}{2} [5 \cdot 3 \cos^2(\theta) (-\sin(\theta)) + 3 \sin(\theta)] \right]$$

Both have r^2 terms so at $r=0$ these vanish.

